

Unit 9 - Two- and Three-Dimensional Shapes

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The following B.E.S.T Standards will be covered in this unit:

MA.912.GR.4.1 Identify the shapes of two-dimensional cross-sections of three-dimensional figures.

- *Clarification 1: Instruction includes the use of manipulatives and models to visualize cross-sections.*
- *Clarification 2: Instruction focuses on cross-sections of right cylinders, right prisms, right pyramids and right cones that are parallel or perpendicular to the base.*

MA.912.GR.4.2 Identify three-dimensional objects generated by rotations of two-dimensional figures.

- *Clarification 1: The axis of rotation must be within the same plane but outside of the given two-dimensional figure.*

MA.912.GR.4.3 Extend previous understanding of scale drawings and scale factors to determine how dilations affect the area of two-dimensional figures and the surface area or volume of three-dimensional figures.

MA.912.GR.4.4 Solve mathematical and real-world problems involving the area of two-dimensional figures.

- *Clarification 1: Instruction includes concepts of population density based on area.*

MA.912.GR.4.5 Solve mathematical and real-world problems involving the volume of three-dimensional figures limited to cylinders, pyramids, prisms, cones and spheres.

- *Clarification 1: Instruction includes concepts of density based on volume.*
- *Clarification 2: Instruction includes using Cavalieri's principle to give informal arguments about the formulas for the volumes of right and non-right cylinders, pyramids, prisms and cones.*

MA.912.GR.4.6 Solve mathematical and real-world problems involving the surface area of three-dimensional figures limited to cylinders, pyramids, prisms, cones and spheres.

Unit 9, Lesson 1: Finding the Area of Two-Dimensional Figures



Benchmark

MA.912.GR.4.4 Solve mathematical and real-world problems involving the area of two-dimensional figures.



Learning Target

- ☐ I can solve mathematical and real-world problems involving the area of two-dimensional figures.



9.1.1 Warm-Up: Michael Jordan

Basketball legend Michael Jordan is often quoted as saying, "I've failed over and over and over again in my life and that is why I succeed."

1. What is a skill, activity, or subject that you have failed at?
2. Explain Jordan's quote in your own words.
3. How can you turn a failure into a success story?



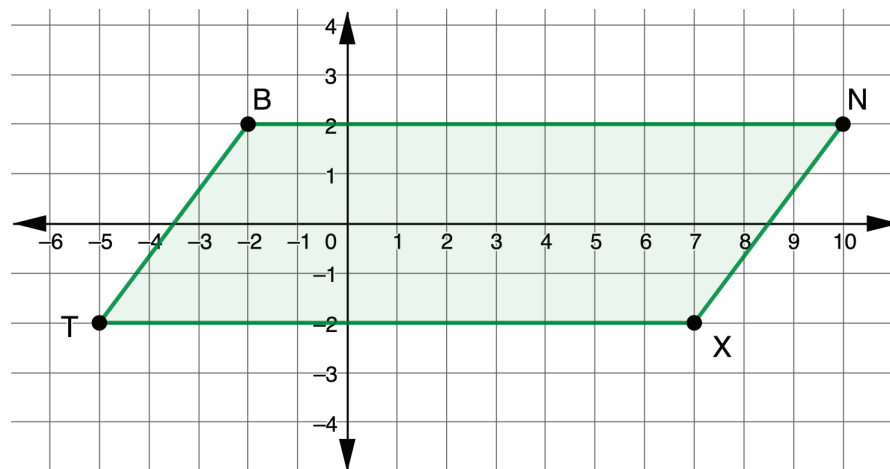
9.1.2 Collaboration: Finding Area

1. Complete the table.

Figure	Area Formula
Rectangle	
Triangle	

2. How is the area of a triangle related to the area of a rectangle?

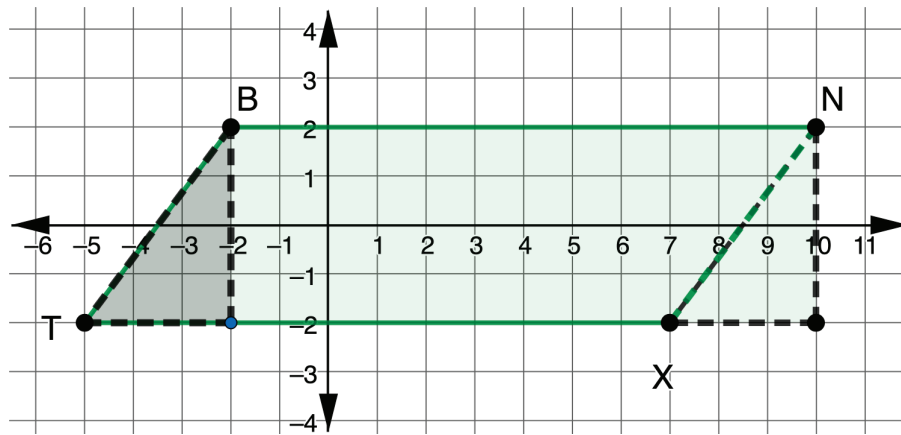
3. Parallelogram $BNXT$ is shown on the coordinate grid.



Tyrese could not remember how to find the area of a parallelogram so he divided the parallelogram into 2 triangles and a rectangle. Work with your partner to use Tyrese's method to find the area of $BNXT$.

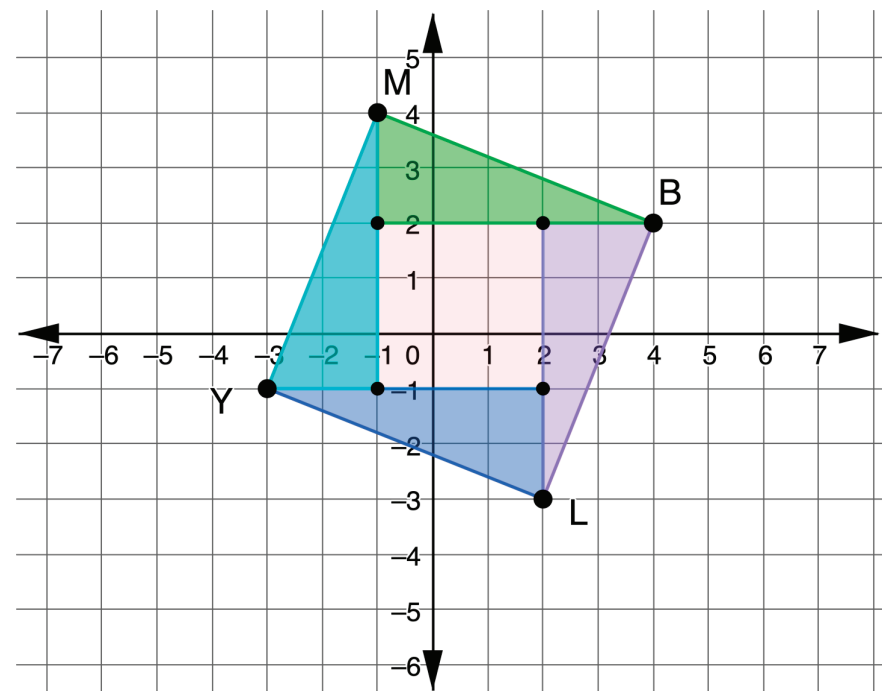
Shape	Area
Triangle 1	
Triangle 2	
Rectangle	
Total Area of Parallelogram	

4. Sarianna rearranged the parallelogram to create a rectangle.

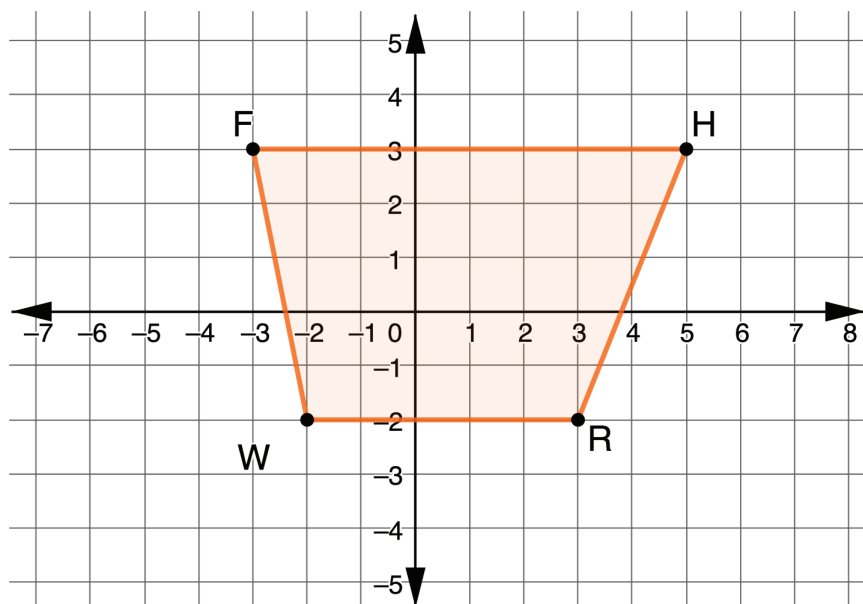


Use Sarianna's rectangle to find the area of $BNXT$.

5. Quadrilateral $MBLY$ is decomposed as shown on the coordinate grid. Find the area of $MBLY$ using the decomposition shown.



6. $FHRW$ is shown on the coordinate grid.



- a. Divide the quadrilateral into two triangles and a rectangle. Then, find the area of each.

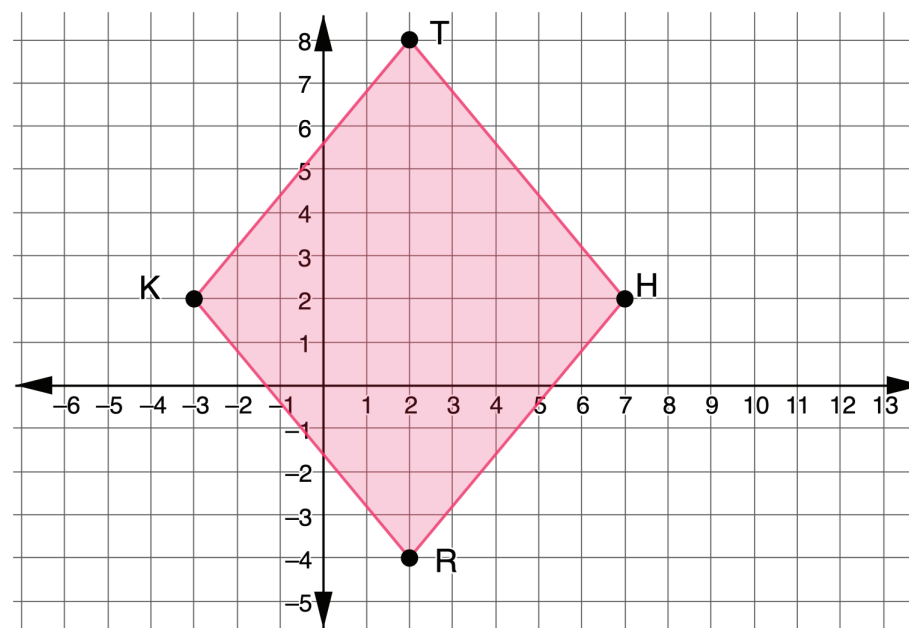
Shape	Area
Triangle 1	
Triangle 2	
Rectangle	

b. What is the area of $FHRW$?

c. What type of quadrilateral is $FHRW$?

7. Ava found a formula for the area of a trapezoid. The formula is $A = \frac{1}{2}h(b_1 + b_2)$, where h is the height and b_1 and b_2 are the bases. Use the formula Ava found to find the area of $FHRW$.

8. $THRK$ is shown on the coordinate grid.



- a. What type of quadrilateral is $THRK$?

- b. Determine the area of $THRK$. Show your work or explain your thinking.

- c. Complete the table.

Line Segment	Length
$\overline{TR}$	
$\overline{KH}$	

The area of a rhombus can be found using the formula $A = \frac{1}{2}pq$, where p and q are the lengths of the diagonals.

- d. Use the formula to find the area of $THRK$.

Formulas for area can easily be forgotten, so it is always helpful to be able to break a figure down into triangles and rectangles in order to find the area. The exception is finding the area of a circle or a part of a circle.

9. What is the area of a circle?

10. The drama program at school is building a new semicircular stage. The longest distance across the stage is 30 feet. What is the area of the new stage? Round to the nearest thousandth.

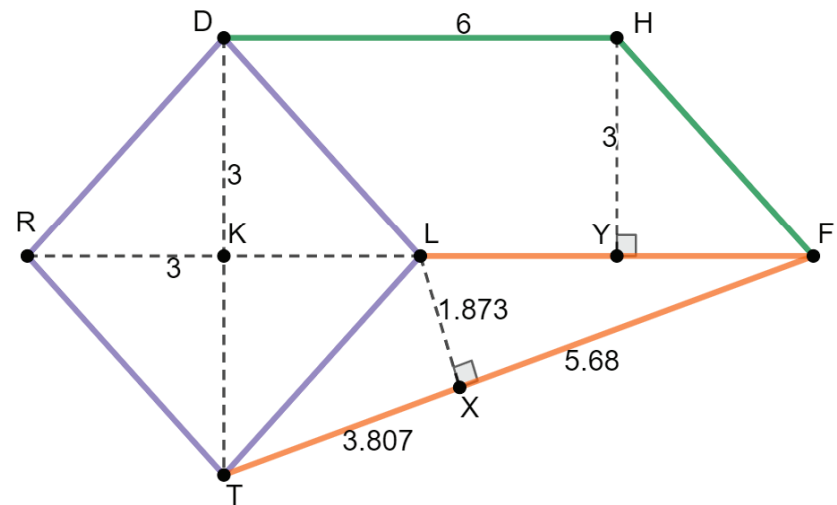


9.1.3 Guided Practice: Using the Formulas

1. The radius of a circle is 7 feet (ft). Find the area, in square feet, of the circle. Leave the answer in π form.
2. The base of the parallelogram is 22 inches (in.). The area of the parallelogram is 132 square inches. Find the height, in inches, of the parallelogram.

3. The diagonals of a rhombus are 40 centimeters (cm) and 18 centimeters. Find the area, in square centimeters, of the rhombus.

4. A composite shape of a rhombus, a parallelogram, and a triangle is shown.



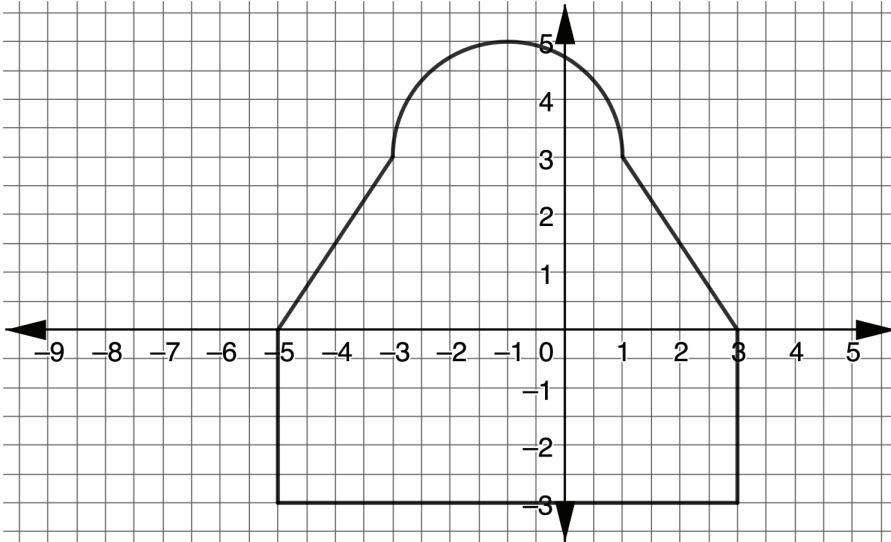
Complete the table. Round to the nearest thousandth.

Shape	Area
Rhombus	
Parallelogram	
Triangle	
Composite Area	



9.1.4 Your Turn!

1. The design of a room is shown, where 1 unit = 5 feet.



Determine the area, in square feet, of the room. Round to the nearest hundredth.



9.1.5 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each.



I need help.



I got it.



I could help someone.

Finding the area of a triangle	  
Finding the area of a rectangle	  
Finding the area of a trapezoid	  
Finding the area of a square	  
Finding the area of a rhombus	  
Finding the area of a parallelogram	  
Finding the area of a circle	  
Finding the area of a composite figure	  

Unit 9, Lesson 2: Density Based on Area



Benchmark

MA.912.GR.4.4 Solve mathematical and real-world problems involving the area of two-dimensional figures.



Learning Target

- ☐ I can solve real-world problems that involve density using area.



9.2.1 Warm-Up: Would You Rather?

- Which car would you rather buy?

Car A	Car B
\$19,000	\$25,000
12% depreciation per year when you upgrade the car after 5 years.	16% depreciation per year when you upgrade the car after 5 years.

- Explain your answer.



9.2.2 Collaboration: Population Density

1. The five largest cities in Florida and their populations are shown.

City	Population
Jacksonville	890,467
Miami	454,279
Tampa	387,916
Orlando	280,832
St. Petersburg	261,338

The table shows the area of Florida's five largest cities.

City	Area (square miles)
Jacksonville	747
Miami	35.87
Tampa	113.41
Orlando	102.4
St. Petersburg	61.74

What Do You Notice About the Population and the Area?

2. Discuss with your partner your understanding of population density. Describe the population density for the city or town you live in.

3. Make predictions about the population density for Florida's five largest cities.

Largest Population Density	Smallest Population Density

4. Explain your thinking.

5. Ask a classmate for the predictions they made as well as their reasons. Record their cities and write your summary of their reason. *You are the only person who should write in the middle two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

	City	My Summary of Their Reason	Initials
Largest Population Density			
Smallest Population Density			



To find population density, divide the population by the area.

$$\text{Population Density} = \frac{\text{population}}{\text{area}}$$

6. Determine the population density, in persons per square mile (persons/mi²), of the five largest cities in Florida and record them in the table from largest to smallest. Round answers to the nearest whole number.

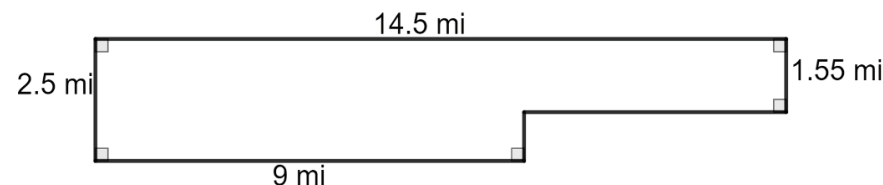
City	Population Density (persons/mi. ²)
1.	
2.	
3.	
4.	
5.	

7. On July 1, 2019, Florida had a population of 21,477,737. The area of Florida is 53,624.76 square miles. Find the population density of Florida.



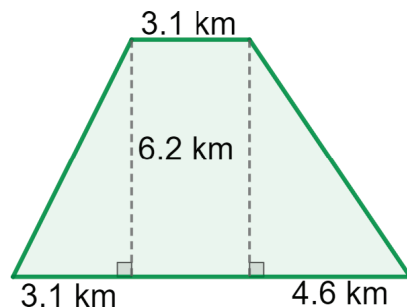
9.2.3 Guided Instruction: Area and Density

1. An outline of the city of Miramar, Florida, with measurements in miles (mi), is shown.



- a. What is the area of the city, in square miles?
- b. Determine the population density of Miramar for 1995 and 2015. Include units.
- | Year | Population | Population Density |
|------|------------|--------------------|
| 1995 | 48,099 | |
| 2015 | 136,996 | |
- c. How did the population density change between 1995 to 2015?

2. The Florida Fish and Wildlife Conservation Commission (FWC) tracks panthers in Florida. By tracking a panther, FWC can map the area in which a panther lives. The habitat of "Florida Panther 251" is shown with dimensions in kilometers, km.



- a. What is the area, in square kilometers (km^2), of Florida Panther 251's habitat?

At one time, the entire southeastern United States was the habitat of the Florida panther. Today, their habitat is restricted to several national parks and wildlife preserves in southwest Florida. Today, the total combined area of the Florida panther habitat is $9,466 \text{ km}^2$.

- b. Based on the area of Florida Panther 251's habitat, how many Florida panthers could be living in southwest Florida?

3. Town A has 20 city blocks, and each city block measures $\frac{1}{20}$ of a mile by $\frac{1}{2}$ mile. There are 6,500 people in this town. Find the population density of the town.



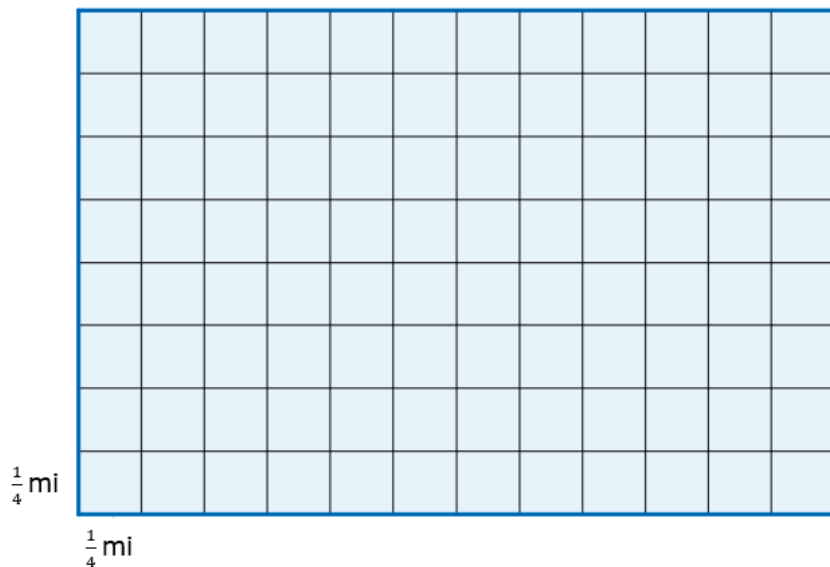
9.2.4 Guided Practice: Density Based on Area

1. The number of leaves on a tree can be estimated using the crown of the tree. A tree crown is the top part of the tree where branches grow out from the main trunk. The width of the tree's crown is used as the diameter of the circle of the ground that lies under the tree's crown. The area where the leaves could fall is 4 times the area of the circle that lays under the tree's crown. The approximate number of leaves per square foot is based on the average size of a leaf for each type of tree. Use the table to approximate the number of leaves that could fall from each tree.

Tree	Crown (in feet)	Approximate Number of Leaves per Square Foot
Sycamore	35	5
Live Oak	16	14

2. The population density of Gainesville, Florida, is 2028.4 persons per square mile, and the population of Gainesville is 124,504. What is the area, in square miles, of Gainesville?

3. Wildlife biologists divided a habitat in the Bear Island area of the Big Cypress National Preserve into 96 squares, as shown. Each square has the dimensions of $\frac{1}{4}$ mile (mi) by $\frac{1}{4}$ mi.



The biologists found that in the Bear Island habitat, there are 13.8 deer per square mile.

- What is the population density for one of the $\frac{1}{4}$ mi by $\frac{1}{4}$ mi square?
- How many deer live in the habitat area the biologists studied?



9.2.5 Wrap-Up: Cool Down

- Write an explanation to a classmate who might have missed today's lesson, in your own words, about population density.

Unit 9, Lesson 3: Cross Sections



Benchmark

MA.912.GR.4.1 Identify the shapes of two-dimensional cross-sections of three-dimensional figures.



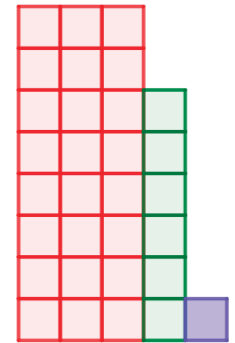
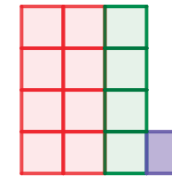
Learning Target

- ☐ I can identify the shapes of two-dimensional cross sections of three-dimensional figures.



9.3.1 Warm-Up: Patterns

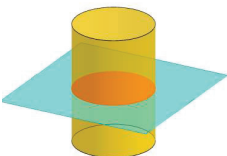
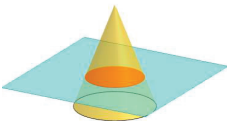
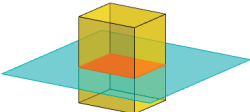
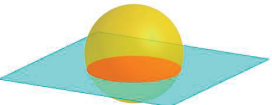
- How many red squares are in the seventh pattern?





9.3.2 Collaboration: Cross Sections

The table below shows the resulting cross section for each type of slice.

Figure	Cross Section	
	Horizontal Plane	Vertical Plane
Cylinder		
Cone		
Rectangular Prism		
Triangular Prism		
Sphere		

1. What do you notice about the diagrams for horizontal slices?
2. What do you notice about the diagrams for vertical slices?
3. Discuss with your partner whether or not the shape of a cross section will change if the height of the plane is changed. Summarize your discussion.
4. Complete the table by describing the shape of the cross section that is created when the figure is sliced by each.

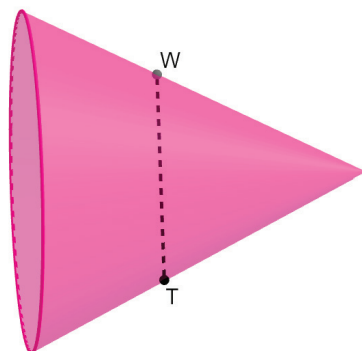
Figure	Cross Section Shape	
	Horizontal Plane	Vertical Plane
Cylinder		
Cone		
Rectangular Prism		
Triangular Prism		
Sphere		



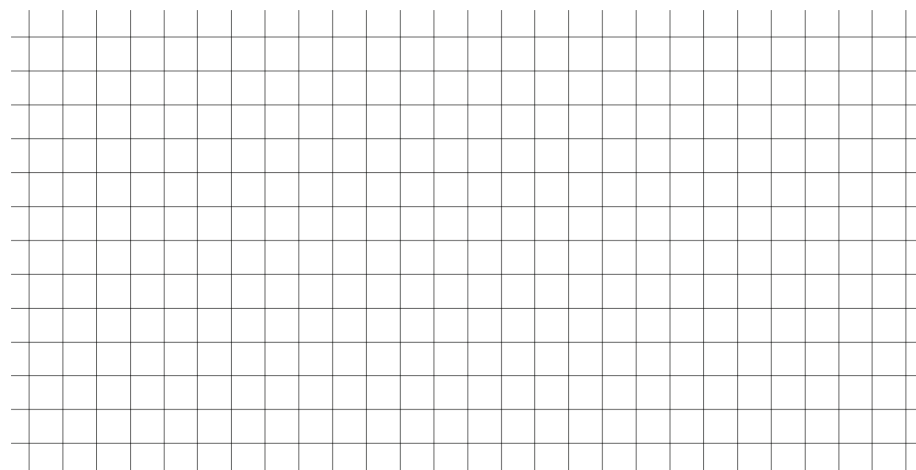
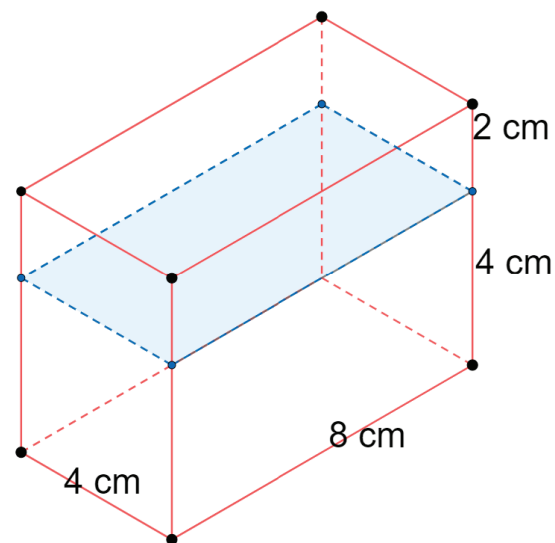
9.3.3 Guided Instruction: Drawing Cross Sections

Three-dimensional figures can be oriented in different ways. Cross sections are typically described by their relation to the base of the three-dimensional figure. For example, cross sections are often described as parallel or perpendicular to the base of the figure.

1. A cone is shown with a cross section parallel to the base. Draw the cross section of the cone on the grid, given $WT = 6$ units.



2. A rectangular prism is shown with a cross section parallel to the base. The dimensions are given in centimeters (cm). Draw the cross section on the grid and label its dimensions.



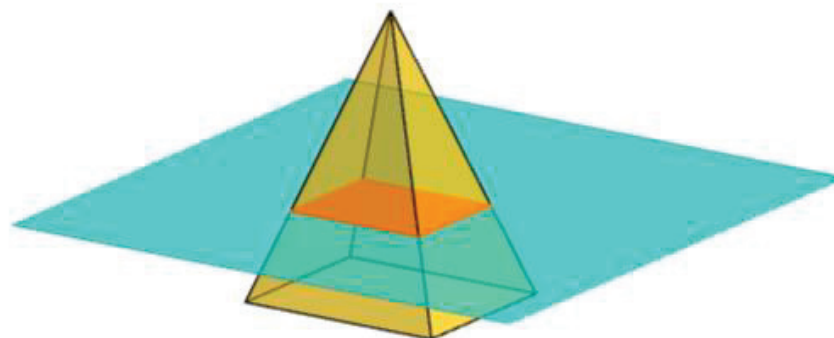
3. A sphere with radius of 4 inches (in.) with a horizontal cross section is shown.



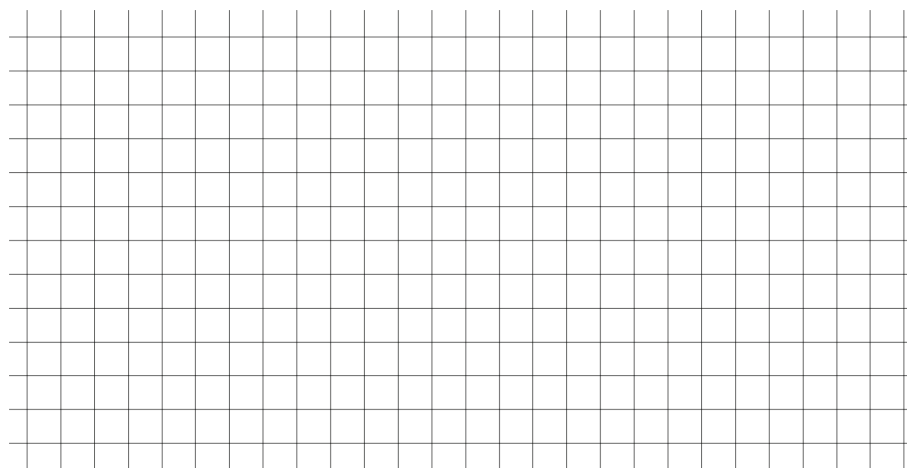
Draw the cross section if it occurs $\frac{3}{8}$ of the distance from the sphere's center.



4. A rectangular pyramid is shown, where the dimensions of the base are 12 centimeters (cm) by 21 cm and the pyramid's height is 33 cm.



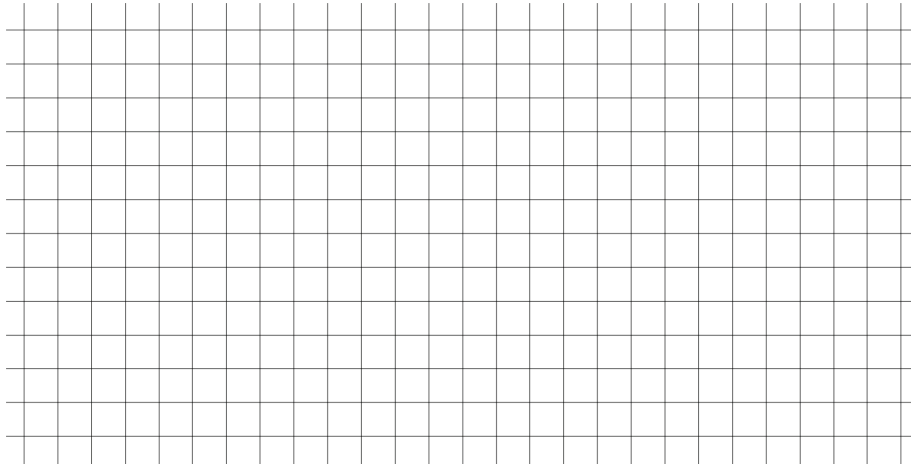
Draw the cross section parallel to the base that is located at a height of 11 cm.





9.3.4 Your Turn!

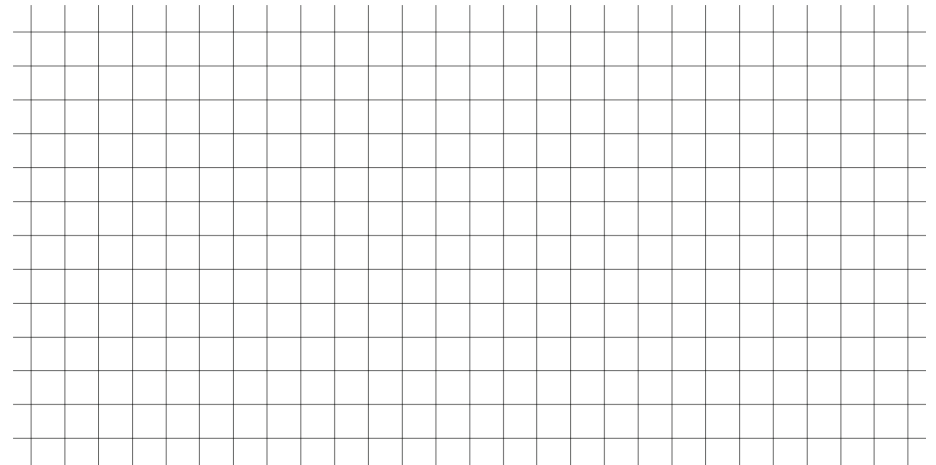
1. Draw the cross section formed by the intersection of a sphere with a diameter of 15 millimeters and a perpendicular plane passing through the center of the sphere.



2. A cylinder has a radius of 4 inches (in.) and a height of 15 in. Draw the cross section formed by the intersection of a cylinder and a plane perpendicular to the base passing through the center of the cylinder.



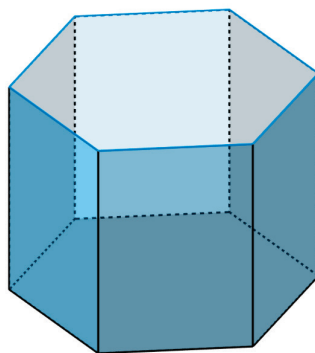
3. Draw the cross section formed by the intersection of a pentagonal prism and a plane parallel to the base passing through the prism.





9.3.5 Wrap-Up: Ticket Out the Door

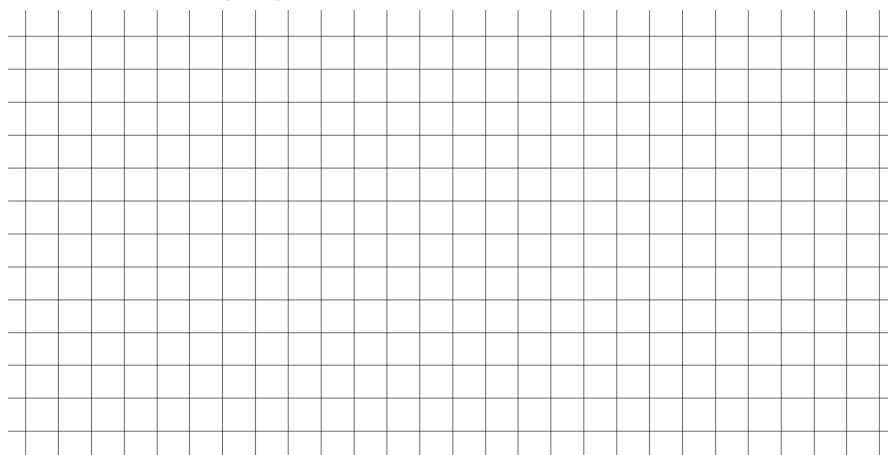
1. A hexagonal prism is shown.



- a. Draw the cross section that results from a plane that is parallel to the base.



- b. Draw the cross section that results from a plane that is perpendicular to the base.



Unit 9, Lesson 4: Solids of Rotation



Benchmark

MA.912.GR.4.2 Identify three-dimensional objects generated by rotations of two-dimensional figures.



Learning Target

- ☐ I can identify three-dimensional objects generated by rotations of two-dimensional figures.



9.4.1 Warm-Up: Notice and Wonder

A Rubik's Cube and a Slinky are shown.



1. What do you notice?
2. What do you wonder?
3. Sketch the horizontal cross section of the Rubik's cube.
4. Sketch the horizontal cross section of the Slinky.



9.4.2 Activity: Rotating Pencils

Work with your partner to complete the following.

1. Place a sticky note on the side of the pencil. If necessary, use tape to secure the sticky note.
2. Make a prediction about which three-dimensional shape the sticky note would form when the pencil is rotated 360° .

Complete the table.

Sticky Note	Prediction	Reasoning

3. Using your pencil as a vertical axis of rotation, rotate the pencil 360° .
4. What three-dimensional figure did the sticky note seem to create?

5. Work with your partner to predict what three-dimensional shape will be formed when each shape is rotated 360° .

Shape	Prediction	Reasoning
Rectangle		
Triangle		
Semicircle		

6. Using your pencil as a vertical axis of rotation, rotate the pencil 360° . Then, sketch or describe the resulting three-dimensional shape.

Shape	Three-Dimensional Shape
Rectangle	
Triangle	
Semicircle	

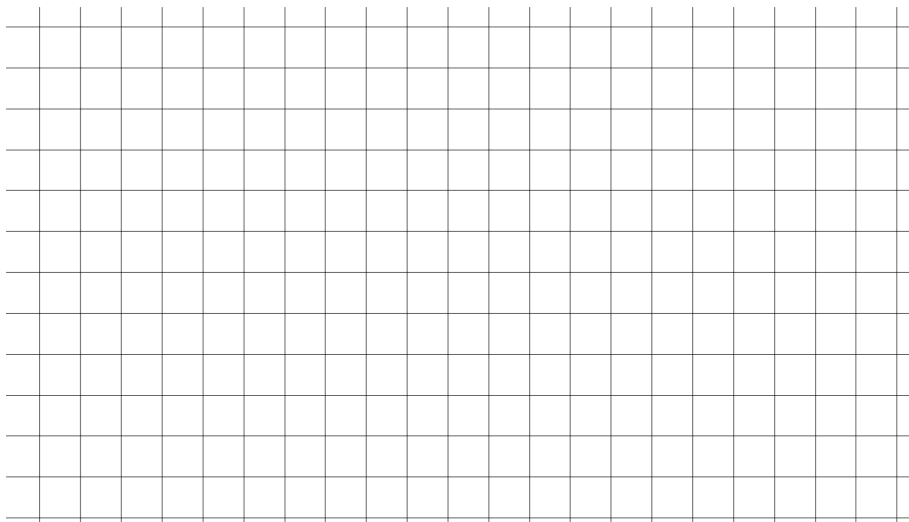
7. Discuss with your partner whether the three-dimensional shape would be different if the pencil were held horizontally instead of vertically. Summarize your discussion.

8. Draw any four-sided, two-dimensional shape on a 3x5 card. Make it as unique as possible. Cut out your shape. Draw the outline of your shape below.

9. Sketch what you think the three-dimensional figure of your four-sided two-dimensional shape will look like if rotated around each axis.

Horizontal Axis	Vertical Axis

10. Review your partner's shape. Then, complete the following.
- Draw your partner's shape on the grid below.
 - Draw a horizontal axis of rotation.
 - Draw the reflection of your partner's shape.



11. Discuss with your partner how drawing a reflection could help envision the three-dimensional figure. Summarize your discussion.

12. Sketch what you think the three-dimensional figure of your partner's four-sided, two-dimensional shape will look like if rotated around each axis.

Horizontal Axis	Vertical Axis

13. Compare your sketches with your partner. Discuss any differences and make any adjustments.

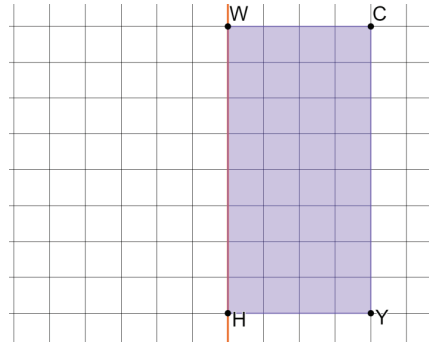
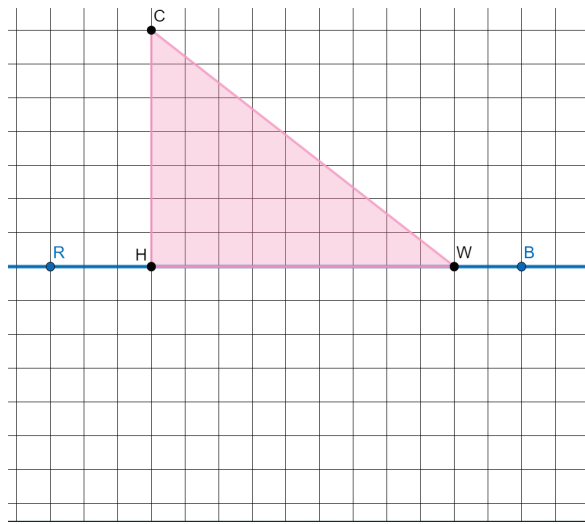


9.4.3 Guided Instruction: Rotations with Dimensions

1. Draw a two-dimensional figure and an axis of rotation that would result in the real-world object when the two-dimensional figure is rotated around the axis of rotation.

Image	Two-Dimensional Figure
A set of revolving doors with a radius of 3 feet. 	
An ice cream cone with a diameter of 4.5 centimeters (cm) and a height of 11 cm. 	

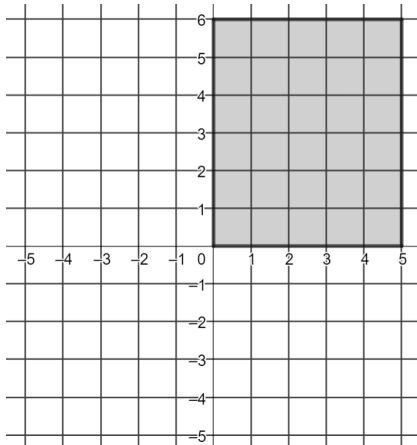
2. Sketch the three-dimensional shape that is created. Include any dimensions in units.

Two-Dimensional Shape	Three-Dimensional Shape
$WCYH$ is rotated around $\overleftrightarrow{WH}$. 	
$\triangle CWH$ is rotated around $\overleftrightarrow{RB}$. 	

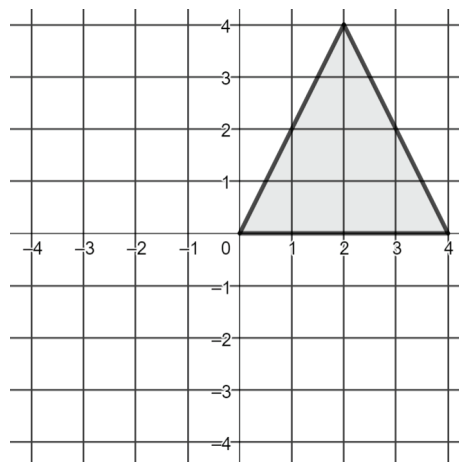


9.4.4 Guided Practice: Rotations

1. Describe and sketch the three-dimensional shape that is created when the rectangle is rotated around the y -axis. Include any known dimensions.

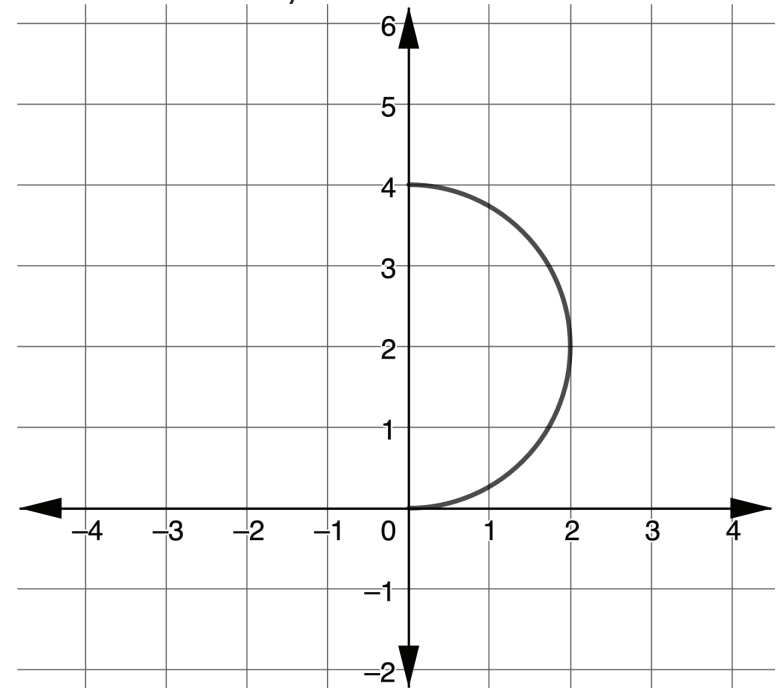


2. Describe and sketch the three-dimensional shape that is created when the isosceles triangle is rotated around the y -axis. Include any known dimensions.



9.4.5 Wrap-Up: Ticket Out the Door

1. Describe and sketch the three-dimensional shape that is created when the semicircle is rotated around its diameter. Include any known dimensions.



Unit 9, Lesson 5: Cavalieri's Principle



Benchmark

MA.912.GR.4.5 Solve mathematical and real-world problems involving the volume of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres.



Learning Target

- ☐ I can use Cavalieri's principle to give informal arguments about the formulas for the volumes of right and non-right cylinders, pyramids, prisms, and cones.



9.5.1 Warm-Up: Notice and Wonder

Four different perspectives of the same stack of records are shown.



1. What do you notice?
2. What do you wonder?



9.5.2 Collaboration: Exploring Cavalieri's Principle

Scenario 1

Consider the two stacks of pennies, where each stack contains the same number of pennies. Discuss with your partner the relationship between the volumes of each stack.

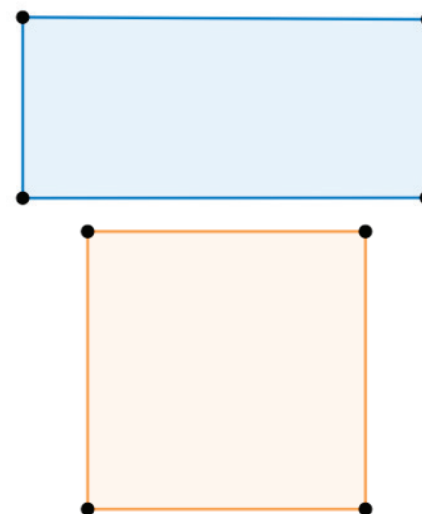


1. Summarize your discussion in the table below and have your partner initial that your summary is correct.

Summary	Partner's Initials

Scenario 2

Suppose two partners are both constructing towers that are to be 5 inches high. Partner A decided to build a tower using rectangles with areas of 16 square inches and partner B decided to construct a tower using squares with areas of 16 square inches. The base of each partner's tower is shown.

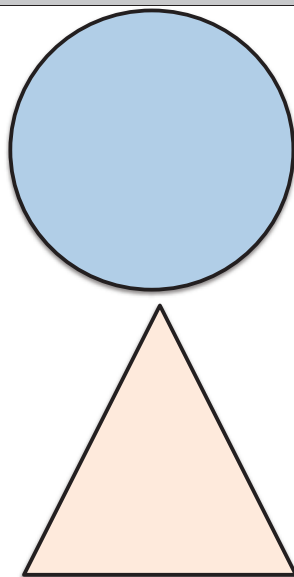


2. Discuss with your partner how the volumes will compare. Summarize your findings in the box below and have your partner initial that your summary is correct.

Summary	Partner's Initials

Scenario 3

ABC ice cream shop decided to run a special for the week and they're offering two serving options for you to choose from. Option 1 is a cone with a circular base that has an area of 4π square inches and a height of 4 inches. Option 2 is a triangular pyramid container with a base of 4π square inches and a height of 4 inches. The base of each option is shown.



- Discuss with your partner how the volumes will compare to one another. Summarize your findings in the box below and have your partner initial that your summary is correct.

Summary	Partner's Initials



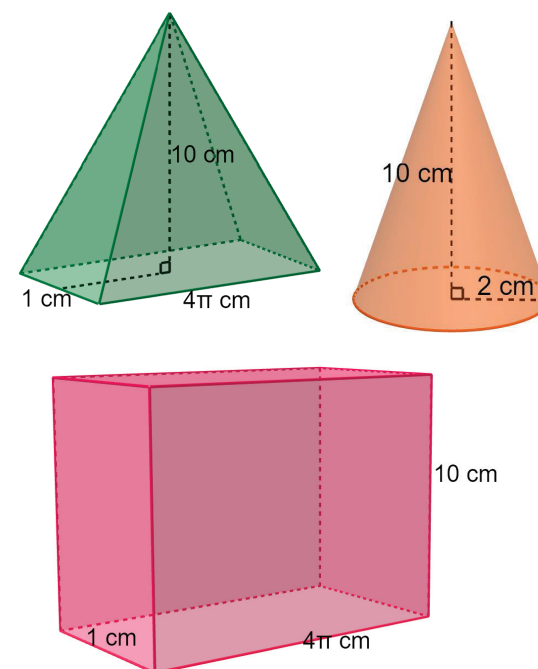
9.5.3 Guided Instruction: Cavalieri's Principle



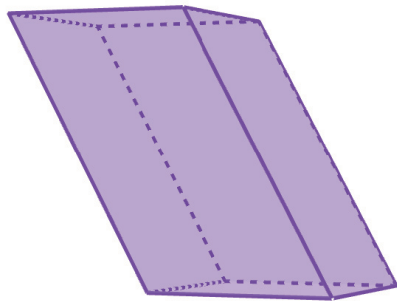
Cavalieri's Principle for Three-Dimensional Solids

If every plane parallel to the two bases of two solids results in cross sections of equal area and the two solids have congruent altitudes, then the solids have equal volumes.

- Using Cavalieri's Principle, determine whether or not the following solids have the same volumes.



2. An oblique prism is shown.

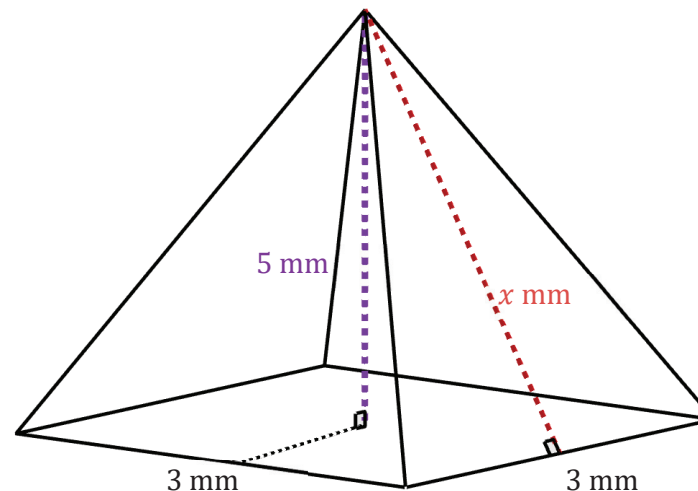


- What shape is each cross-section of the oblique prism?
- What is the area formula for the base?
- What is the volume formula for an oblique rectangular prism?
- Discuss with your partner how the formula for the oblique rectangular prism compares to the formula for a right rectangular prism. Explain your reasoning.



9.5.4 Guided Instruction: Slant Height vs. Vertical Height

A square pyramid is shown with measurements in millimeters (mm).



Samantha and Julian each found the measure of the slant height, x , by first drawing a diagram and then applying the Pythagorean Theorem. Their work is shown.

Samantha	Julian
$c^2 = 3^2 + 5^2$	$c^2 = 1.5^2 + 5^2$

1. Discuss with your partner which student correctly set up their work to determine the slant height. Summarize your discussion.



The **slant height** of an object is the distance from the base to the apex along a lateral face, whereas the **vertical height or altitude** of an object is the perpendicular distance from the apex to the base.

2. Discuss with your partner which height is important when finding the volume of an oblique figure. Summarize your discussion.



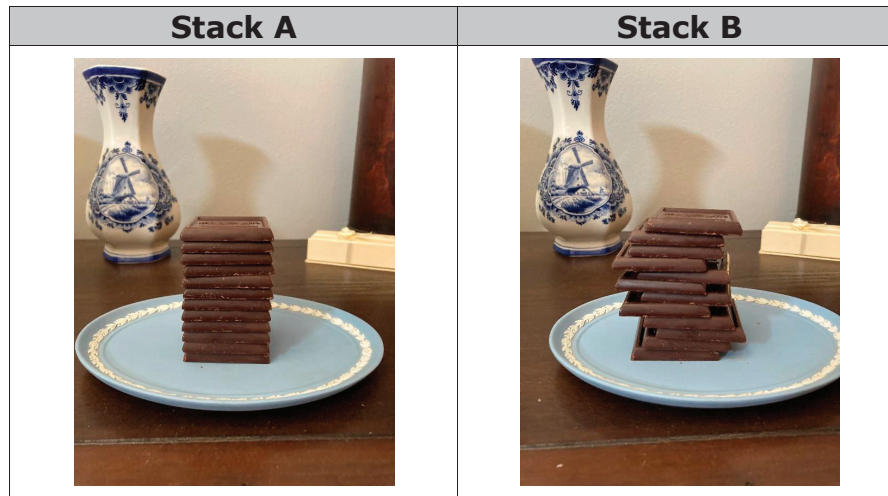
9.5.5 Guided Practice: Cavalieri's Principle in the Real World

1. Consider the stack of sticky note pads below.



If each pad is a square with side lengths of 3 inches and a height of 0.97 inches, approximate the volume of a stack of 8 sticky note pads.

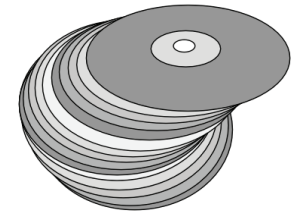
2. Two stacks of 12 chocolate squares are shown.



Each chocolate square has the dimensions $1\frac{5}{8}$ inches (in.) by $1\frac{5}{8}$ in. Each square is $\frac{1}{8}$ in. thick. Find the volume of each stack.

Stack A	Stack B

3. While cleaning out an attic, you found an old stack of compact discs (CDs) that were used to store music. If each disc has an area of 36π cm², explain how you can find the approximate volume of a stack of 25 CDs.



4. A zigzag crystal vase is shown, where the vase has a height of 10 inches (in.). The cross-sections that are parallel to the base are always rectangles that are 6 in. wide by 3 in. long.



- Determine the volume of the vase, assuming the crystal itself has no thickness.
- The crystal is actually half an inch thick on each of the sides and on the bottom. Approximately how much space is contained within the vase? Explain or show your reasoning.



9.5.6 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.



Benchmark

MA.912.GR.4.5 Solve mathematical and real-world problems involving the volume of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres.



Learning Target

- ☐ I can find the volume of spheres, prisms, cones, pyramids, and cylinders.



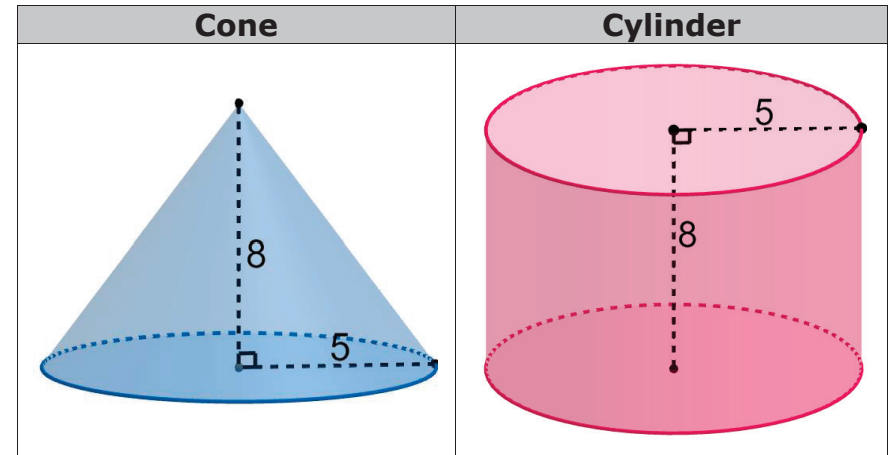
9.6.1 Warm-Up: Would You Rather?

1. Would you rather have a pool that is 20 meters (m) by 8 m by 6 m, or a pool that is 1550 centimeters (cm) by 650 cm by 725 cm?
 - a. Use mathematics to compare the two pools.
 - b. Which pool would you rather have?



9.6.2 Guided Instruction: Finding Volume

1. A cone and cylinder are shown, where the radius of the base of each is 5 units and the height of each is 8 units.

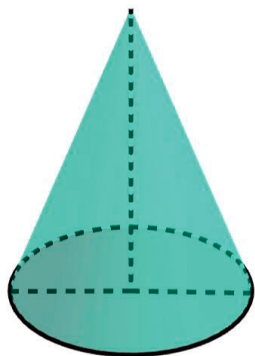


- a. Suppose the cone is filled with water. Discuss with your partner how many cones you think you would need to fill the cylinder with water. Write your estimate.
- b. Given the volume of a cylinder is $V = Bh$, discuss with your partner how to re-write the formula to find the volume of the cone.
- c. The volume formula for a cone is $V = \underline{\hspace{2cm}}$



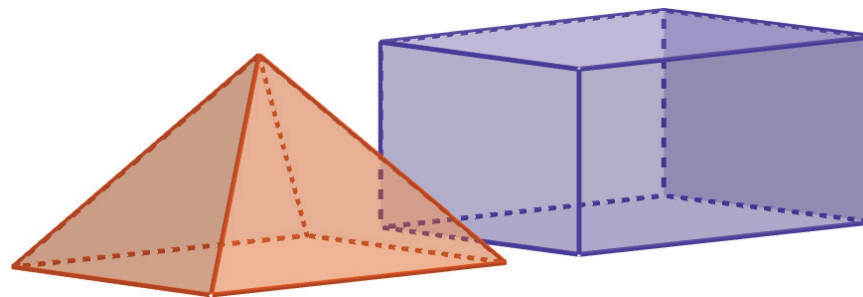
A **cone** is a three-dimensional figure with a circular base and an apex that is connected to the base by a collection of line segments that form a curved surface.

2. A cone is shown with a diameter of 16 millimeters (mm), a height of 22 mm, and a slant height of 23.4 mm.



Determine the volume of the cone, in terms of π .

3. The bases of the rectangular pyramid and rectangular prism shown have congruent bases, and each has the same height.

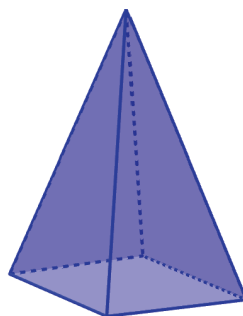


- Suppose the rectangular pyramid is filled with water. Discuss with your partner how many rectangular pyramids you think you would need to fill the rectangular prism with water. Write your estimate.
- Given the volume of a prism is $V = B \cdot h$, discuss with your partner how to re-write the formula to find the volume of the pyramid.
- The volume formula for a regular pyramid is $V = \underline{\hspace{2cm}}$



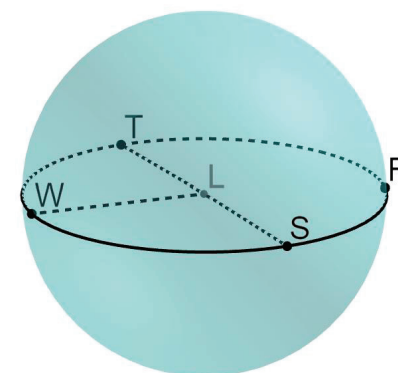
A **pyramid** is a figure with a polygonal base and triangular faces. The triangular faces have the same size and shape and they connect the sides of the base to a common point called the apex.

4. A pyramid is shown, whose base is a rectangle with dimensions of 10 centimeters (cm) by 14 cm. The height of the pyramid is 17 cm. Determine the volume of the pyramid.



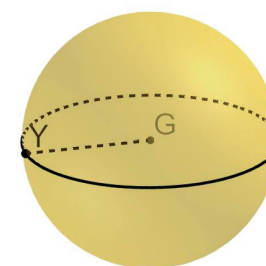
A **sphere** is a three-dimensional figure, where all points are equidistant from a point called the center.

A sphere with center L is shown, where $\overline{LW}$ is a radius and $\overline{TS}$ is a diameter. The circle that passes through the points $TRSW$ is known as a great circle since it is the largest possible circle that can be made around the sphere.



The volume formula for a sphere is $V = \frac{4}{3}\pi r^3$.

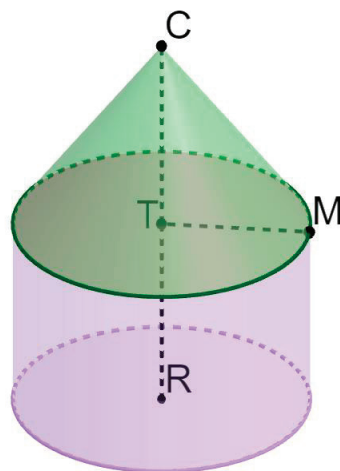
5. Find the volume of the sphere G , where $GY = 3$ using $\pi \approx 3.14$. Round to the nearest hundredth.



6. Composite figures are made up of more than one figure. The composite figure shown is a cone on top of a cylinder.

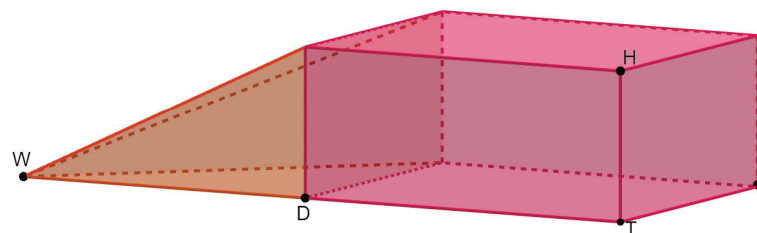
Check all that must be true about the composite figure.

- ☐ The heights have the same measure.
- ☐ The diameters have the same measure.
- ☐ The radii have the same measure.
- ☐ The bases have the same area.
- ☐ The volume of this composite figure can be found using the formula $V = 2\pi r^2 h$.



7. Discuss with your partner the statements that were false. Rewrite each false statement so that it would be true.

8. A composite figure is shown, where $HT = 3.2$, $KT = 4.8$, $DT = 10.7$, and $WD = 11.1$. WD is the height of the pyramid. Find the volume of the composite figure.





9.6.3 Guided Practice: Finding Volume

1. Find the volume of a hemisphere that has a diameter of 9.4 inches. Round your answer to the nearest thousandth.
2. A composite figure consists of a hemisphere and a cone. The base of the cone and the great circle of the hemisphere have the same diameter. The radius of the hemisphere is $4\frac{3}{4}$ centimeters (cm). The height of the cone is $9\frac{1}{12}$ centimeters (cm). Find the volume of the composite figure using $\frac{22}{7}$ for π . Round your answer to the nearest thousandth.



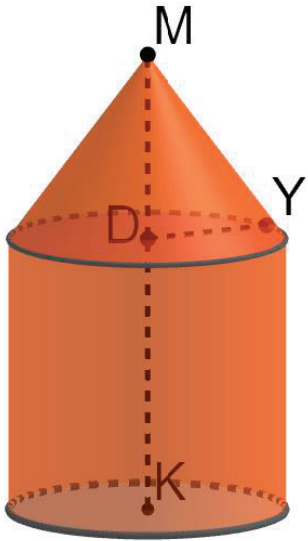
9.6.4 Your Turn!

1. Find the volume of a cylinder with a diameter of 18.3 millimeters (mm) and a height of 6.4 mm. Round your answer to the nearest thousandth.
2. Find the volume of a pyramid whose height is 15.7 inches (in.), and whose base is a rectangle with dimensions of 7.6 in. by 12.4 in. Round your answer to the nearest thousandth.



9.6.5 Wrap-Up: Ticket Out the Door

1. Find the volume of the composite figure if $MD = 17$, $DY = 8.5$, and $KD = 26$. Round your answer to the nearest hundredth.



Unit 9, Lesson 7: Volume in the Real World



Benchmark

MA.912.GR.4.5 Solve mathematical and real-world problems involving the volume of three-dimensional figures limited to cylinders, pyramids, prisms, cones and spheres.



Learning Target

- ☐ I can use volume to solve problems with real-world applications.



9.7.1 Warm-Up: Math Fail

- The clearance sign shown is what is known as a “math fail.”

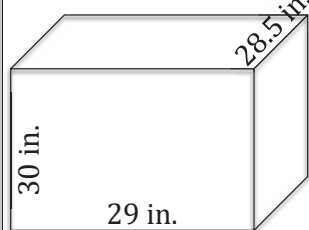
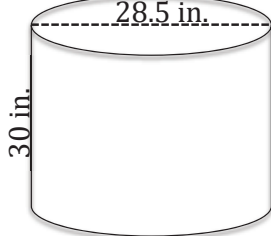
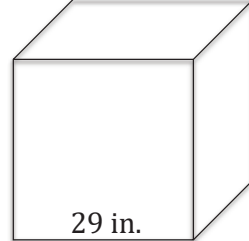


- What do you notice about the sign that makes this a math fail?
- Write a letter to the company explaining why this is a math fail and how to fix the issue.



9.7.2 Collaboration: Packing Peanuts

- Josiah works for a sporting goods company in the shipping department. Josiah is trying to determine the best solution for packing a basketball with a diameter of 28.5 inches (in.).
 - What is the volume of the basketball?
 - What size box do you think would be the best fit?
 - Find the volume of the three options of boxes below, from which Josiah can choose.

Rectangular Prism		Volume = _____
Cylinder		Volume = _____
Cube		Volume = _____

- d. Discuss with your partner how to determine the amount of space left over in the package after the basketball has been placed in it. Summarize your discussion.

- e. After the basketball has been placed in the package, the remaining space is filled with packing peanuts. If each peanut is 1 cm^3 , which box will use the least amount of packing peanuts?



9.7.3 Guided Practice: Volume

- Keon is building a raised garden bed that will have a width of 3.5 feet (ft), a depth of 2 ft, and length of 3 ft. If Keon fills the garden bed with soil, how much soil does he need to purchase?
- Jasper wants to buy a new coffee mug, and he has narrowed it down to two options. The dimensions are given in inches (in.).

Option A	Option B
 5.25 in Diameter: 2 in	 5 in Radius: 1.25 in

- What is the volume, rounded to the thousandths place, of option A?
- What is the volume, rounded to the thousandths place, of option B?



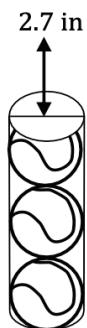
9.7.4 Your Turn!

1. A tennis ball canister with three tennis balls is shown. The diameter of the canister is 2.7 inches (in.) and the height is 8.1 in.

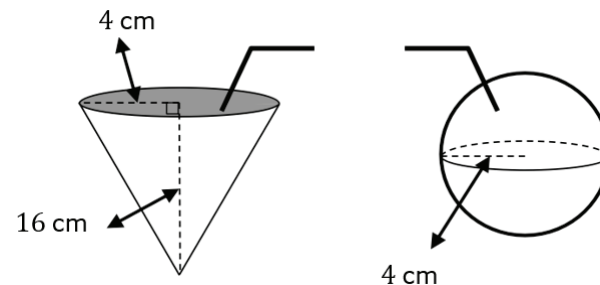
a. What is the volume of the canister?

b. What is the volume of 1 tennis ball?

c. How much air is left in the canister after the tennis balls have been placed inside?



2. Club Cool in Epcot offers free samples of Coke products that are distributed worldwide. Flavors include Fanta Pineapple, Fanta Melon, Inca Cola, VegitaBeta, and Sparberry. You can choose to have your sample served in either of the below containers shown.



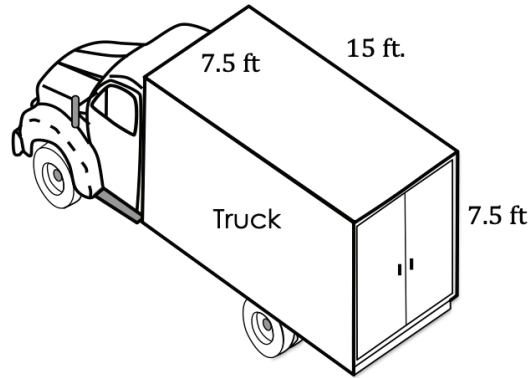
Gian, Yanessa, and Jack made different conclusions about the drink containers. Who is correct and why?

Gian: The volumes are the same because the height of the cone is 4 times its radius, and both containers have the same radius.

Yanessa: The volume of the sphere is greater than the cone's because $\frac{4}{3}\pi r^3$ is greater than $\frac{1}{3}\pi r^2 h$.

Jack: The volume of the cone is greater than the sphere's because the height is 4 times the radius of both containers.

3. Julian owns a moving company and has a truck that measures $15\text{ ft} \times 7.5\text{ ft} \times 7.5\text{ ft}$.



He charges individuals by the number of boxes in the moving truck. If each box is a cube with side lengths of 1.5 ft, how many boxes can he fit in the truck?

4. The Pyramid of Cestius, pictured below, is located in Rome, Italy. The base of the pyramid is a square with side lengths of 30 meters and stands at a height of 36.5 meters. What is the volume of the pyramid?





9.7.5 Wrap-Up: Cool Down

1. Provide a real-world context that would require you to find the volume of a three-dimensional figure.



Unit 9, Lesson 8: Surface Area of Three-Dimensional Figures



Benchmark

MA.912.GR.4.6 Solve mathematical and real-world problems involving the surface area of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres.



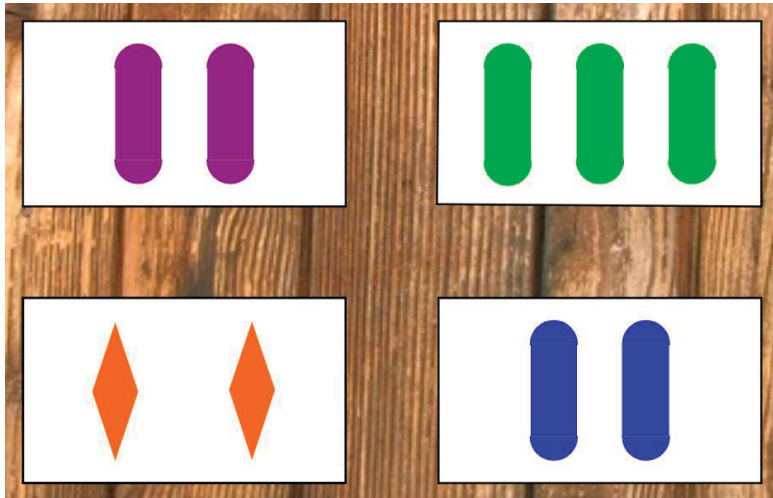
Learning Targets

- ☐ I can solve mathematical problems involving the surface area of cylinders, pyramids, prisms, cones, and spheres.
- ☐ I can solve mathematical problems involving the lateral area of cylinders, pyramids, prisms, cones, and spheres.



9.8.1 Warm-Up: Which One Doesn't Belong?

- Which one doesn't belong? Explain your answer.



9.8.2 Collaboration: Using Nets to Find Surface Area



The **lateral area** of a three-dimensional figure is the sum of the area of the figure's faces.



The **surface area** of a three-dimensional figure is the sum of the lateral area of the figure's faces and the area of the figure's bases.

- Work with your partner to determine a formula for the lateral area and the surface area of the figure you have been given. Complete the table.

Our figure was:

Dimension				
Variable Used				



9.8.3 Activity: Gallery Walk

- During the gallery walk, record the formulas. Make any useful notes on each figure's net.

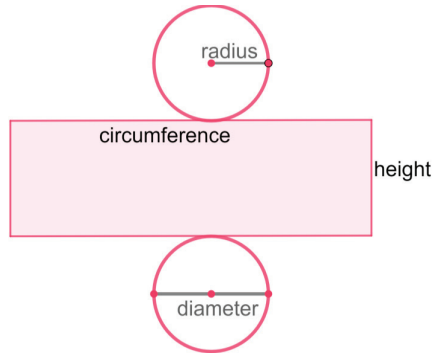
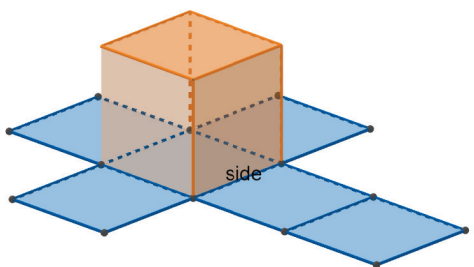
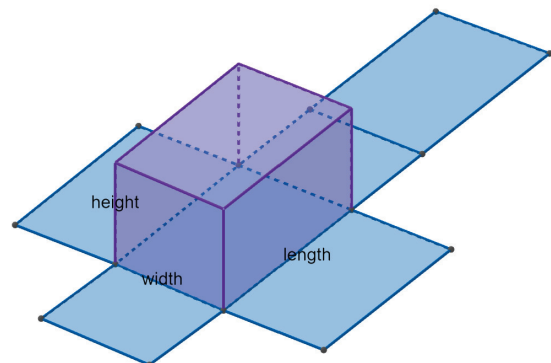
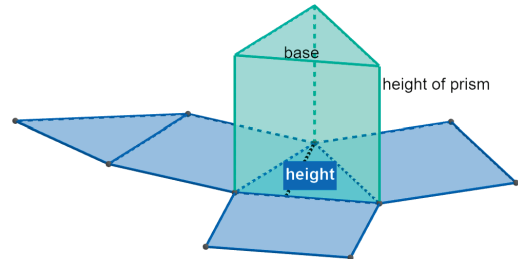
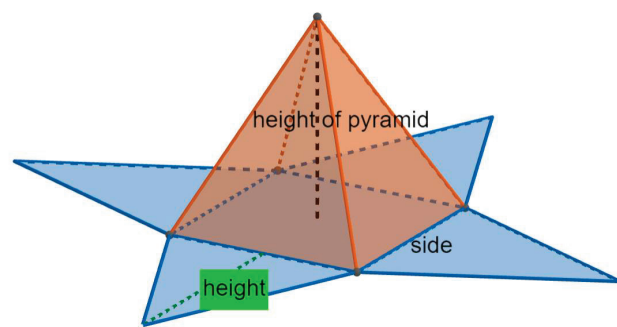
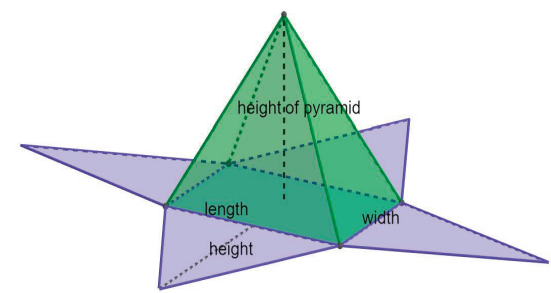
Figure	Net
Cylinder	
Lateral Area =	
Surface Area =	
Cube	
Lateral Area =	
Surface Area =	
Rectangular Prism	
Lateral Area =	
Surface Area =	

Figure	Net
Triangular Prism	
Lateral Area =	
Surface Area =	
Square Pyramid	 The height of the triangle that is a face of a pyramid is also known as slant height, h_s.
Lateral Area =	
Surface Area =	
Rectangular Pyramid	 The height of the triangle that is a face of a pyramid is also known as slant height, h_s.
Lateral Area =	
Surface Area =	



9.8.4 Bring It Together: Formulas

Using the net of three-dimensional objects can help to find the lateral area and surface area. The net of a cone and a sphere make deriving the formulas for surface area from the net more challenging. A chart of formulas is given, where SA represents surface area.

Figure	Surface Area	Variables
Prism	$SA = 2B + Ph$	B = base, P = perimeter, h = height
Cylinder	$SA = 2B + Ch$ Or $SA = 2B + Ph$	B = base, C = circumference (or P = perimeter, h = height)
Cone	$SA = B + \pi r h_s$	B = base, r = radius, h_s = slant height
Regular Pyramid (a pyramid whose base is a regular polygon)	$SA = B + \frac{1}{2}Ph_s$	B = base, P = perimeter, h_s = slant height
Sphere	$SA = 4\pi r^2$	r = radius

1. The formulas in the table may not match the formulas recorded during the gallery walk. Which figures have formulas that are different?

2. There are many ways to write and to understand formulas. Discuss with your partner which formulas make more sense to you. Discuss how the formulas in the formula chart can be used to help you remember how to find surface area. Summarize your discussion.

3. The chart does not have a formula for lateral area. For each of the figures, except for spheres, circle the part of the surface area formula that gives the lateral area.



9.8.5 Guided Instruction: Lateral Area and Surface Area

1. A square pyramid has a slant height of 7 inches, and the area of the base is 16 square inches. What is the surface area, in square inches?
2. The surface area of a sphere is 144π feet. What is the radius of the sphere?
3. If a cylinder has a height of 16 centimeters and a radius of 8 centimeters.
 - a. Find the lateral area. Leave your answer in π form.
 - b. Find the surface area. Leave your answer in π form.
4. A right triangular prism has base side lengths of 8, 15, and 17 feet (ft), where 17 ft is the hypotenuse of the triangle. The prism has a height of 7 ft. What is the surface area, in square feet, of the triangular prism?



9.8.6 Guided Practice: Lateral Area and Surface Area

1. A pyramid has a square base whose area is 49 square feet with a height of 11.5 feet. Find the lateral area, in square feet, of the pyramid.
2. A rectangular prism has a length of 12.5 inches (in.), a width of 8 in., and a height of 5 in. Determine the surface area, in square inches, of the rectangular prism.
3. The radius of a sphere is 18.7 feet. Find, in square feet, the surface area of the sphere. Round your answer to thousandths place.



9.8.7 Your Turn!

1. Find the lateral area, in square meters, of a cone that has a radius of 3 meters (m) and a slant height of 8.5 m. Round your answer to thousandths place.

2. A cylinder has a radius of 5 inches (in.) and a height of 3.75 in. Find the surface area, in inches, of the cylinder. Round your answer to thousandths place.



9.8.8 Wrap-Up: Error Analysis

1. Cody and Mila are finding the surface area of a pyramid whose base area is 144 inches and slant height is 16 inches. Their work is shown in the table.

Cody's Work	Mila's Work
$SA = B + \frac{1}{2}Ph_s$	$SA = B + \frac{1}{2}Ph_s$
$SA = 144 + \frac{1}{2}(48)(16)$	$SA = (144) + \frac{1}{2}(144)(16)$
$SA = 528$ square inches	$SA = 1296$ square inches

- a. Whose work is correct?

- b. Write a note to the student whose work is incorrect describing the mistake the student made.

Unit 9, Lesson 9: Surface Area in the Real World



Benchmark

MA.912.GR.4.6 Solve mathematical and real-world problems involving the surface area of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres.



Learning Targets

- ☐ I can solve real-world problems involving the surface area of cylinders, pyramids, prisms, cones, and spheres.
- ☐ I can solve real-world problems involving the surface area of composite figures.



9.9.1 Warm-Up: Math Magic

Arthur Benjamin (b. 1961) is a professor of mathematics at Harvey Mudd College in California. Dr. Benjamin is often referred to as a “mathemagician” because he is known for his quick mental computational skills. Dr. Benjamin’s TED talks have been viewed millions of times on the internet.

Try one of Dr. Benjamin’s math “magic tricks.”

- Pick a number between 1 and 10.
- Double your number.
- Add 10 to your number.
- Divide by 2.
- Subtract the number you started with originally.

1. What number did you get?

2. Compare your number with your partner’s number. What do you notice?

3. Why do you think the “trick” works?

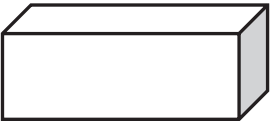
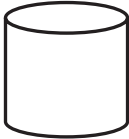
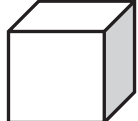




9.9.2 Collaboration: The Best Package

Manuel is a graphic designer and works with various types of packaging. For the artwork, Manuel needs a container with a surface area of at least 250 square inches (in.) to ship the company's products.

1. The three types of packaging that the company has in stock are shown in the table. Find the surface area for each type of packaging.

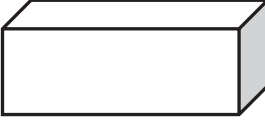
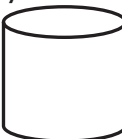
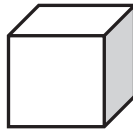
Packaging	Dimensions	Surface Area
Rectangular Prism 	l : 7 in. w : 10 in. h : 4 in.	
Cylinder 	d : 8 in. h : 8 in.	
Cube 	s : 12 in.	

2. Rank the packaging's surface area from smallest to largest.

Rank	Packaging
1 (smallest)	
2	
3 (largest)	



3. The prices for packaging are listed in the table. Based on the surface area of each package, find the cost of each type of packaging.

Packaging	Pricing	Cost of Packaging
Rectangular Prism 	\$1.75 per 100 square feet.	
Cylinder 	\$2.25 per 100 square feet.	
Cube 	\$1.25 per 100 square feet.	

4. If Manuel wants to use the cheapest packaging possible for his client, which packaging should he choose?



9.9.3 Guided Instruction: Real-Life Applications of Surface Area

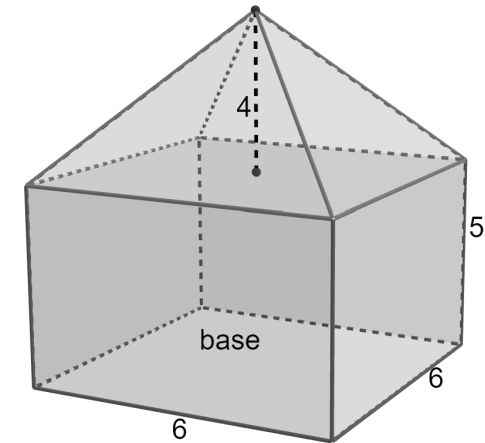
Jeni owns a bakery and is making two cakes.

Cake A	Cake B
Cake A is in the shape of a rectangular prism that is 3 inches thick, 9 inches long, and 12 inches wide.	The cake is in the shape of a cylinder and has a diameter of 8 inches and a height of 6 inches.

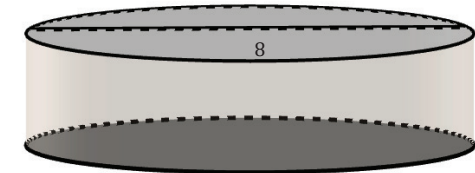
1. When Jeni frosts the cake, she applies the frosting to all surfaces except for the bottom of the cake. Determine the amount of frosting Jeni needs to apply to each cake. Show your work.

Cake A	Cake B

2. Jeni makes a two-layer cake, as shown, where the measurements are in inches (in.) She will frost all of the cake except for the base. The frosting between the pyramid and the prism helps to hold the two cakes together. Determine the total surface area that Jeni needs to frost.



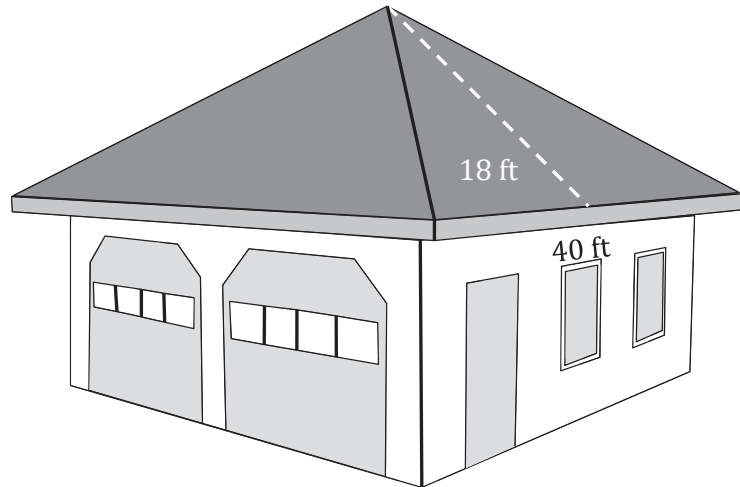
3. Jeni's recipe for buttercream icing typically frosts an 8-inch round cake that is 1.5 inches high. How many batches of frosting will Jeni need to frost the two-layer cake?





9.9.4 Guided Practice: Real-Life Applications of Surface Area

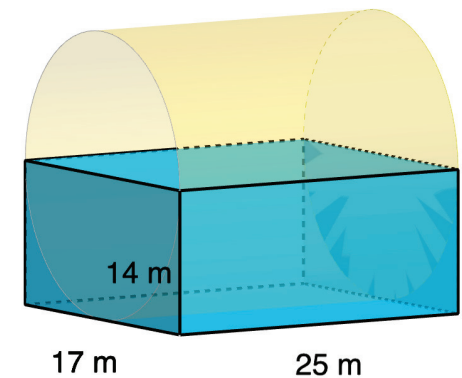
1. Lourdes' garage is in the shape of a square pyramid. The garage is shown, with dimensions given in feet (ft).



Lourdes is replacing the roof and installing shingles. In roofing terminology, a "square" is 100 square feet. Composition shingles come in bundles that cover $\frac{1}{3}$ of a square.

How many bundles should Lourdes buy to cover the roof?

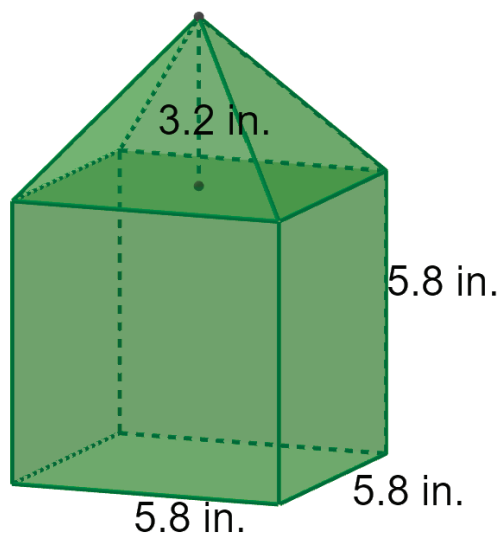
2. An architect is designing an art gallery. She used computer software to create a rough sketch of her idea. Her sketch is shown with measurements in meters (m). She would like for the art gallery to be all glass. To determine the estimated cost, she would need to find the area of all of the surfaces except for the floor. Determine the total surface area, in square meters, of the glass in the art gallery. Round to the nearest thousandth.





9.9.5 Wrap-Up: Ticket Out the Door

1. Jillian was building a birdhouse. Some of the dimensions, in inches (in.), of the birdhouse are shown. Determine the total amount of wood, in square inches, that Jillian will need to build the outside of the birdhouse.



Unit 9, Lesson 10: Determining How Dilations Affect Area



Benchmark

MA.912.GR.4.3 Extend previous understanding of scale drawings and scale factors to determine how dilations affect the area of two-dimensional figures and the surface area or volume of three-dimensional figures.



Learning Target

- ☐ I can discover the relationship between area and the scale factor of a dilation.



9.10.1 Warm-Up: Where is Here?

A picture of a directional sign is shown.

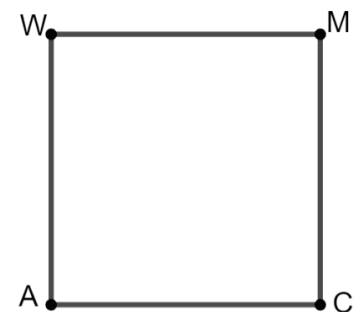


1. What do you notice about the sign?
2. Where do you think the picture was taken?
3. The Earth's average polar circumference is 24,859 miles. What do you think the sign means?



9.10.2 Exploration: Scale Factor and Area

1. Square $AWMC$ is shown, where $WM = 1$ unit.



- a. Determine the area of $AWMC$.
- b. $AWMC$ is dilated using point A as the center of dilation using each of the given scale factors. Complete the table.

Scale Factor	$\frac{1}{2}$	$\frac{3}{4}$	2	3	4
Side Length of Dilation					
Area of Dilated Square					

- c. Write a conjecture about the relationship between the length of the scale factor and the value of the area.

- d. Determine the area of $AWMC$ if $WM = 4$ units.

2. Suppose square $DTSC$ is dilated using point D as the center of dilation for each of the given scale factors. Complete the table for $DTSC$ given that $TS = 4$ units.

Scale Factor	$\frac{1}{2}$	$\frac{3}{4}$	2	3	4
Side Length of Dilation					
Area of Dilated Square					

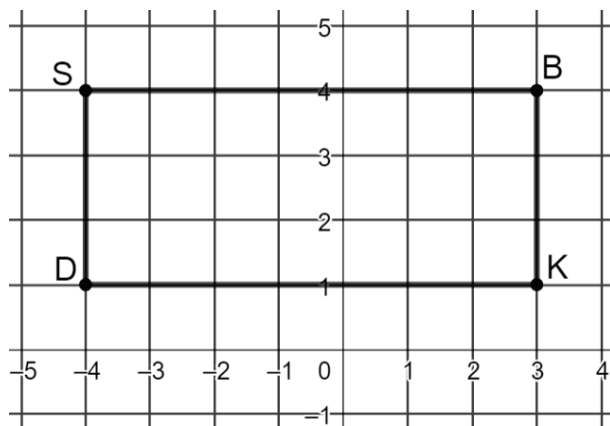
- a. Review your conjecture from the previous problem and make any revisions, if needed.

- b. Share your conjecture with your partner. Listen to your partner's conjecture and write a summary. Have your partner initial that it is summarized correctly.

Summary of Partner's Conjecture	Partner's Initials

- c. Review your conjecture and make any revisions, if needed.

3. Rectangle $SBKD$ is shown.



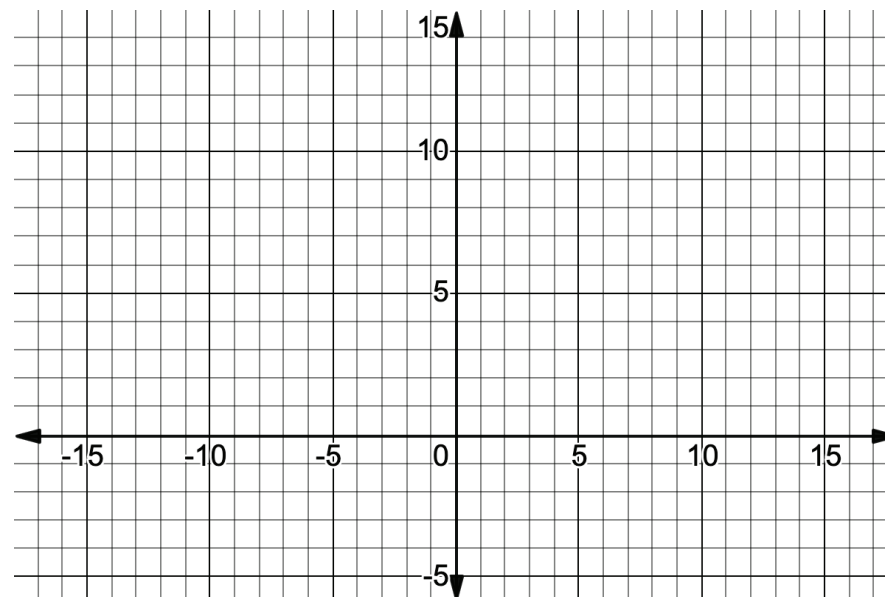
- a. Using the coordinate grid, find the area of the rectangle.

Area = _____ square units

- b. Dilate $SBKD$ by a scale factor of 3 centered at the origin. Write the new coordinates in the table.

Point	Preimage	Image
S	$(-4, 4)$	
B	$(3, 4)$	
K	$(3, 1)$	
D	$(-4, 1)$	

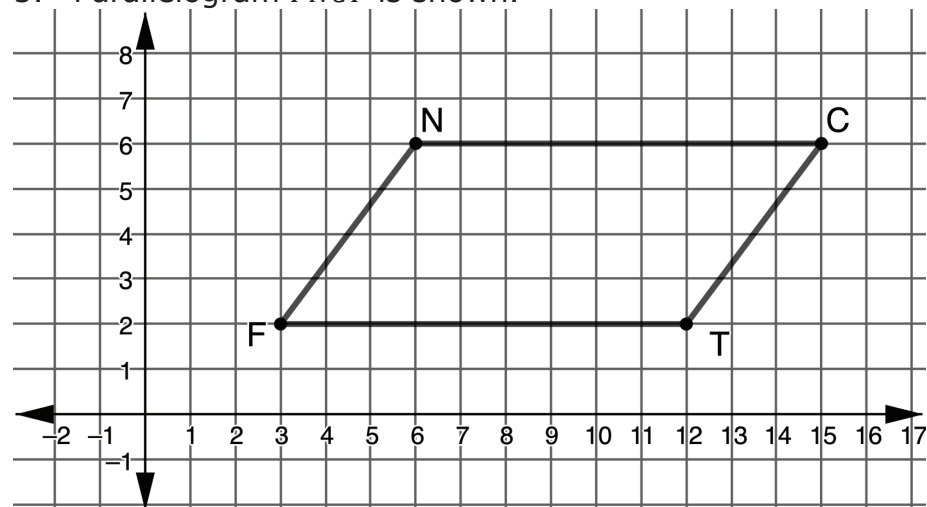
4. Graph $S'B'K'D'$ on the coordinate plane.



- a. The area of rectangle $S'B'K'D'$ is _____ square units.

- b. Does the conjecture you wrote about the relationship between area and scale factor when dilating a square hold true for rectangle $SBKD$?

5. Parallelogram $FNCT$ is shown.



a. What is the area of $FNCT$, in square units?

b. Dilate $FNCT$ by a scale factor of $\frac{2}{3}$ centered at the origin. Write the coordinates of the preimage and image in the table.

Point	Preimage	Image
F		
N		
C		
T		

c. Plot the coordinates of $F'N'C'T'$ on the same coordinate plane as $FNCT$.

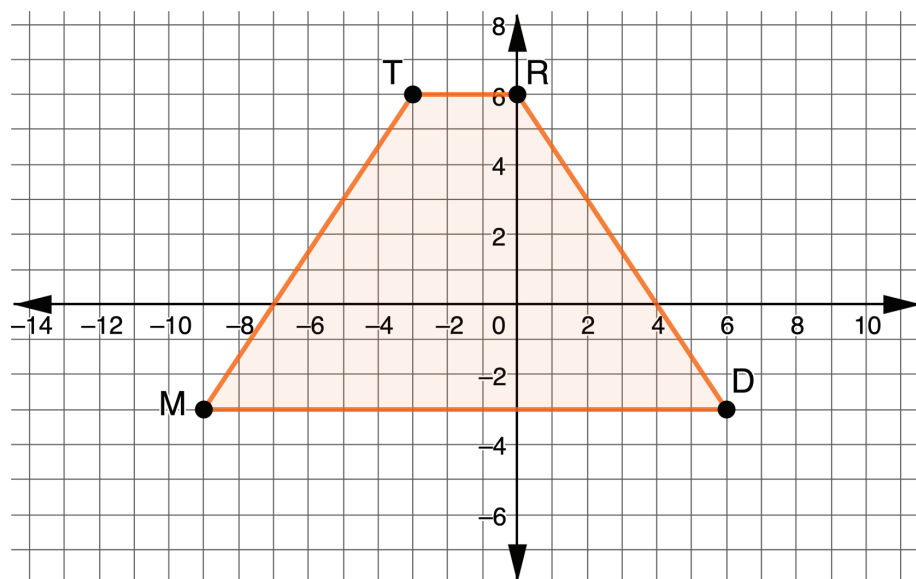
d. What is the relationship between the area of $FNCT$ and $F'N'C'T'$?

e. How can the scale factor of $\frac{2}{3}$ be used to determine the area of $F'N'C'T'$ from the area of $FNCT$?



9.10.3 Guided Practice: Solving Problems Involving Scale Factor and Area

1. Trapezoid $MTRD$ is shown on the coordinate plane.



- a. Find the area of $MTRD$.

$$A = \frac{1}{2}(\text{ } + \text{ })(\text{ })$$

The area of $MTRD$ is _____ square units.

- b. Find the coordinates of $MTRD$ given the scale factors in the table. Use the preimage coordinates to find the coordinates of the images. Both dilations are centered at $(0,0)$.

Point	Preimage	Dilation $k = \frac{1}{6}$ ($M'T'R'D'$)	Dilation $k = \frac{4}{3}$ ($M''T''R''D''$)
M			
T			
R			
D			

- c. Find the area of $M'T'R'D'$.

- d. What effects does the scale factor of $\frac{1}{6}$ have on the original area?

e. Find the area of $M''T''R''D''$.

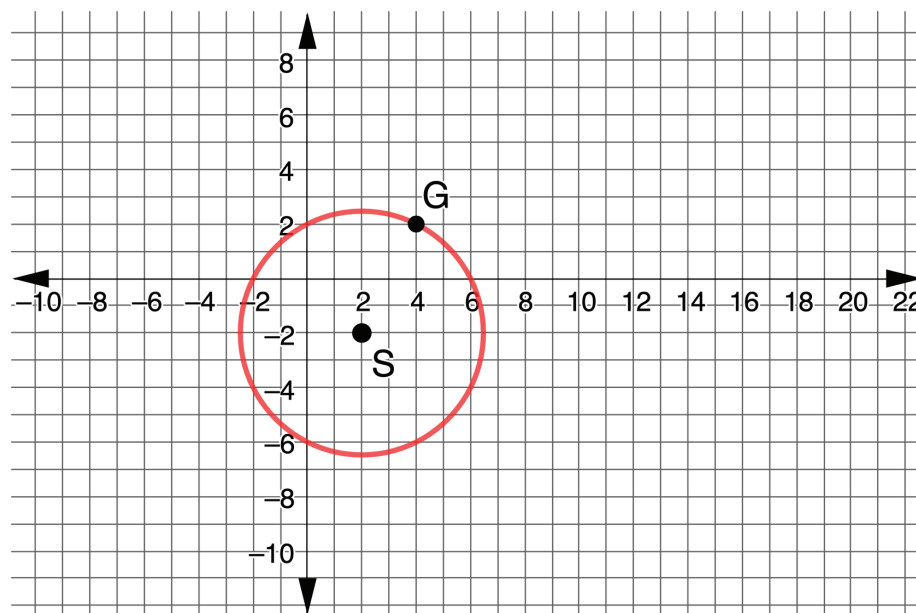
f. What effects does the scale factor of $\frac{4}{3}$ have on the original area?

g. How does the area of the trapezoid with a scale factor of $\frac{1}{6}$ differ from the area of the trapezoid with a scale factor of $\frac{4}{3}$?



9.10.4 Your Turn!

1. Circle S is shown with a radius of $\sqrt{20}$.



a. Dilate circle S using a scale factor of 1.5 centered at the origin and graph the resulting image on the coordinate plane.

b. Find the area of the circles.

Area of Preimage	Area of Image



9.10.5 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about how dilations affect area.

Unit 9, Lesson 11: Determining Surface Area Given Dimension Changes



Benchmark

MA.912.GR.4.3 Extend previous understanding of scale drawings and scale factors to determine how dilations affect the area of two-dimensional figures and the surface area or volume of three-dimensional figures.



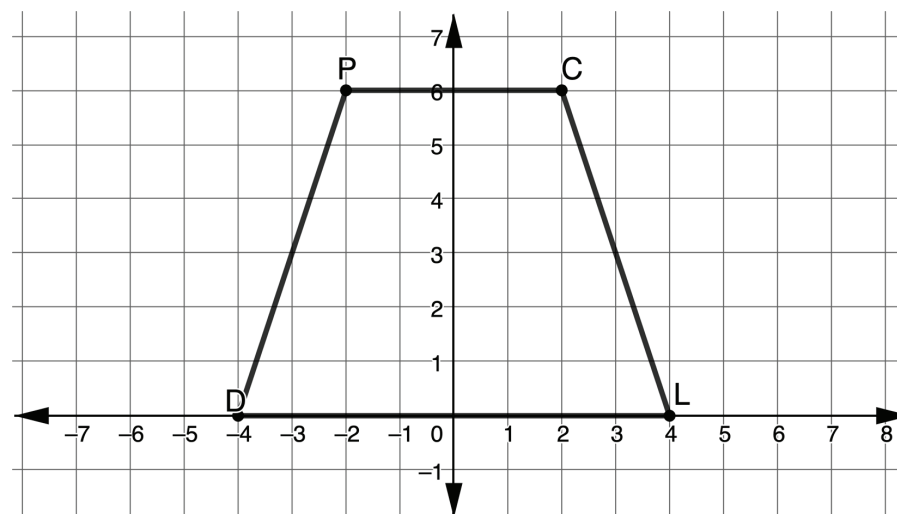
Learning Targets

- ☐ I can find the area of a figure when one or both dimensions are changed.
- ☐ I can find the surface area of a three-dimensional figure when one or both dimensions are changed.



9.11.1 Warm-Up: Bell Work

1. Trapezoid $DPCL$ is shown in the coordinate plane.



- a. Find the area of $DPCL$.
- b. Find the coordinates of the image of $MTRD$ given the scale factor in the table. Use the preimage coordinates to find the coordinates of the image. The dilation is centered at $(0,0)$.

Point	Preimage	Dilation $k = 6$ ($D'P'C'L'$)
D		
P		
C		
L		



9.11.2 Exploration: Ryan's Renovations

Ryan is building a new house and wants to make the size of the kitchen larger. The current kitchen is 24 feet wide and 10 feet long. Ryan wants to make the new kitchen 20 feet long but leave the width the same.

- Find the area of the original kitchen and the area of the new kitchen.

Area of the Original Kitchen	Area of the New Kitchen

- How do the areas compare?

Ryan also wants to install a pool for his new house. The construction company gives Ryan three shape options for the new pool. Each pool's dimensions are given in feet (ft).

Pool 1	Pool 2	Pool 3
Cylindrical	Rectangular Prism	Cube
diameter = 16 ft height = 12 ft	length = 30 ft width = 20 ft height = 12 ft	Each side = 15 ft

- Find the surface area of each of the shapes for the three pool options. Round to the nearest thousandth.

Pool 1	Pool 2	Pool 3

The construction company is giving Ryan more options for building a swimming pool. The three options are listed in the table.

Pool 4	Pool 5	Pool 6
Cylindrical	Rectangular Prism	Cube
diameter = 8 ft height = 6 ft	length = 15 ft width = 10 ft height = 6 ft	Each side = 7.5 ft

- Describe how the dimensions of pair of pools are related.

Pair	Description
Pool 1 and Pool 4	
Pool 2 and Pool 5	
Pool 3 and Pool 6	

5. Find the surface area of each of the shapes for the three new options for a pool. Round to the nearest thousandth.

Pool 4	Pool 5	Pool 6

6. Record the surface area of all pool shapes in the table.

Pool	Surface Area
1	
2	
3	
4	
5	
6	

The cost of building the swimming pool is dependent upon the amount of material. Since the top of each of the shapes would be water, determine the surface area of each pool that could be used to determine the cost.

7. Complete the table.

Pool	Surface Area for Material
1	
2	
3	
4	
5	
6	

8. If Ryan wants the least expensive pool, which one should he purchase?

9. Which pool would you purchase? State your reasoning.



9.11.3 Guided Practice: Ryan's House

Ryan is building a bathroom and needs to install either a bathtub or a shower. The options for each bathroom size are given.

Shower 1	Tub 1
length = 4 ft width = 6 ft height = 12 ft	length = 4 ft width = 8 ft height = 10 ft

1. In order to determine the amount of drywall, Ryan needs to find the surface area of each of the room options.

Shower 1	Tub 1

2. Ryan determines that the dimensions of the shower and tub are not up to code. The updated dimensions are shown. Use the new dimensions to determine the surface area of each room.

Shower 2	Tub 2
length = 6 ft width = 9 ft height = 12 f	length = 6 ft width = 12 ft height = 15 ft
Surface Area	Surface Area

Two of Shower 2's dimensions were changed by a factor of 1.5, whereas all of Tub 2's dimensions were changed by a factor of 1.5. Discuss with your partner on how to use the factor 1.5 and the number of changes in dimensions to find the new surface area from the original surface area.

3. Complete the statements.

_____ $\times$ surface area of Shower 1 = surface area of Shower 2

_____ $\times$ surface area of Tub 1 = surface area of Tub 2



9.11.4 Your Turn!

The living room of Ryan's house is 12 feet long and 18 feet wide.

1. The room is a rectangular shape. Find the area of Ryan's living room.
2. How would the area of the living room change if Ryan decided to enlarge the dimensions by 25%?
3. What is the scale factor for a dimension increase of 25%?
4. How did the area of the living room change with the new dimensions?



9.11.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.